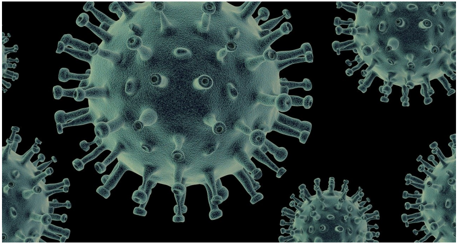
Deep analysis of time series - Example.

19 stycznia 2021



Example - Temperature in Tokyo 1976-2010








Example - Temperature in Tokyo 1976-2010

We study the monthly temperature in Tokyo in 1976-2010. We try to fit the following classical decomposition model:

$$X_t = m_t + s_t + Z_t$$

where

- m_t is a polynomial trend

$$m_t = a_0 + a_1 * t + a_2 * t^2 + a_3 t^3 \quad a_0, a_1, a_2, a_3 \in \mathbb{R};$$

- s_t is the seasonality with period $T = 12$;
- Z_t is a stationary ARMA(p,q) process such that $EZ_t = 0$ and

$$Z_t = \sum_{i=1}^{p} \phi_i Z_{t-i} + \epsilon_t - \sum_{j=1}^{q} \theta_j \epsilon_{t-j}.$$

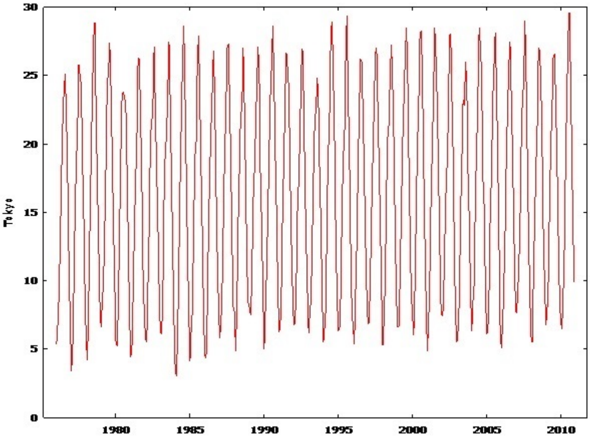
We verify this model.

Procedure

1. Smoothing the data by moving average filter;
2. Estimation of seasonality;
3. Estimation of parameters a_0, a_1, a_2 and a_3;
4. Verification of normality of residuals Z_t;
5. Preliminary estimation of the model of Z_t: ACF and PACF;
6. Final estimation of ARMA(p,q) model of Z_t;
7. Analysis of residuals ϵ_t whether is white noise: ACF, PACF, Pierce-Box and Ljung-Box test.

The data

The month temperture in Tokyo 1976-2010 is presented in the picture.



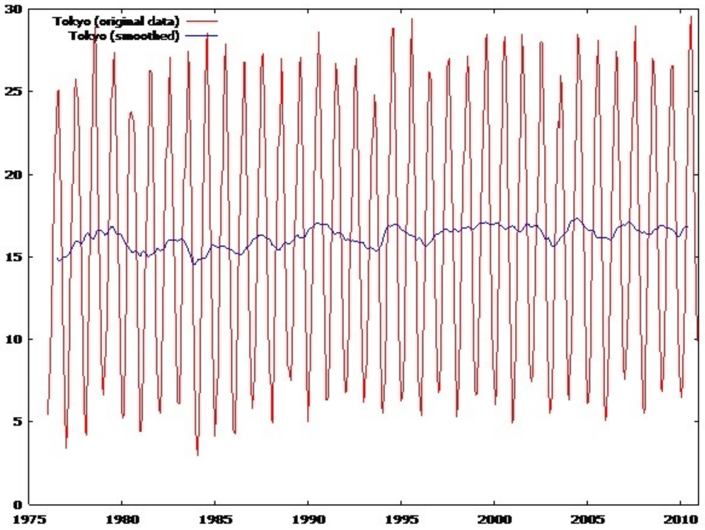
Smoothing

- Let t is the index of month.
- The number of months is $n = 12 * (2010 - 1976) = 408$
- The seasonality is $T = 12$ (every $T = 12$ months the temperature procedure should repeat). Using the smoothing by moving average

$$\hat{m}_t^{MA} = \frac{0.5 * X_{t-6} + \sum_{i=-5}^{5} X_{t+i} + 0.5 * X_{t+6}}{12}$$

for $t = 7, 8, ..., 402$ ($t = 7, ..., n - 6$).
- The plot of smoothed data is on the next page.

The raw and the smooth data



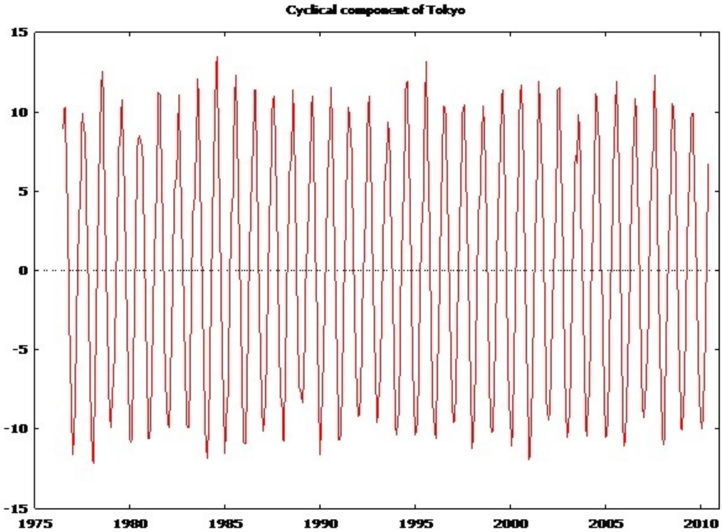
The cyclical component and the seasonality

- The interpretation of the plot provides the preliminary view on the trend.
- It seems the trend is slightly increasing, but the slope is so small we cannot prejoudge.
- Further analysis are needed.
- We try to estimate the seasonality.
- The next slide shows the "cyclical component"

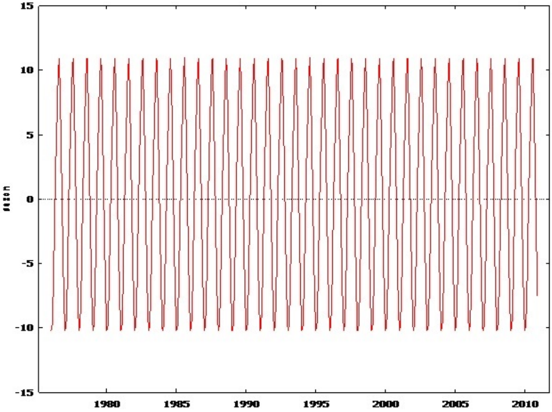
$$\tilde{s}_t = X_t - \hat{m}_t^{MA}$$

as an the data for estimation of seasonality.

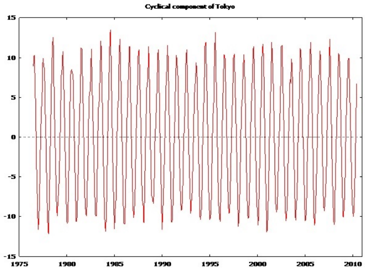
Cyclical component

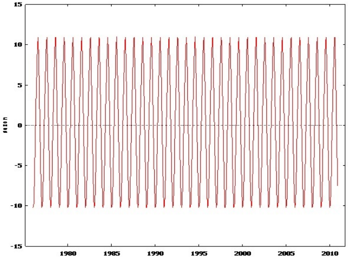


After averaging every the time series Y_t for January (mean of $\tilde{s}_{13}, \tilde{s}_{25}, \ldots, \tilde{s}_{37}$), for February (mean of $\tilde{s}_{14}, \tilde{s}_{26}, \ldots$) etc. we compute the estimator of seasonality $\hat{s}_t$ whose picture is :



Cyclical component ant the seasonality





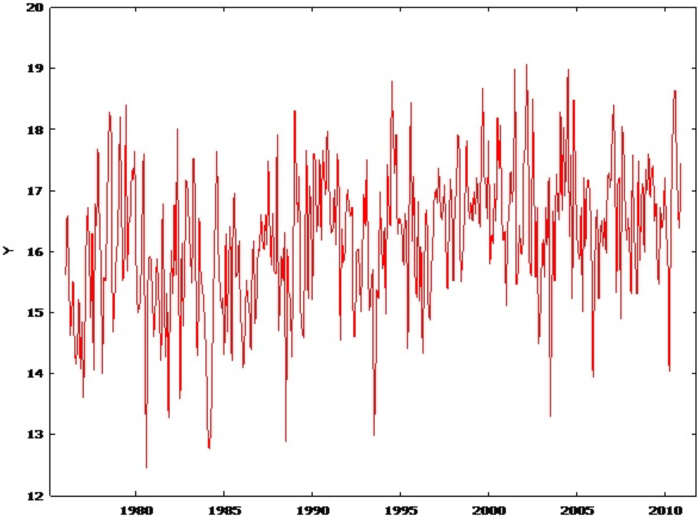
Removing seasonality

- Having the estimator of seasonality $\hat{s}_t$ we subtract it from the raw data and we have

$$Y_t := X_t - \hat{s}_t \approx a_0 + a_1 t + a_2 t^2 + a_3 * t^3 + Z_t.$$

- We find the coefficient of this polynomial by the known *least squared method*.
- The plot of Y_t is on the next figure.

The data after removing the seasonality



The data after removing seasonality

- We can observe that Y_t circulates around the monotone function;
- We try to fit the polynomial of third degree to points (t, Y_t) $(t = 1, 2, ..., n)$.
- Using the least squared method we have the result reported in the next table.

The final estimation

	coefficient	std. error=$S_{\hat{a}_i}$	t-ratio= $\frac{\hat{a}_i}{S_{\hat{a}_i}}$	p-value
a_0	15.6925	0.214867	73.03	$2.35 * 10^{-239} < 0.05$
a_1 (by t)	-0.00309561	0.00441471	-0.7012	$0.4836 > 0.05$
a_2 (by t^2)	$4.15374 * 10^{-05}$	$2.43518 * 10^{-5}$	1.706	$0.0888 > 0.05$
a_3 (by t^3)	$-6.95124 * 10^{-08}$	$3.80250 * 10^{-8}$	-1.828	$0.0683 > 0.05$

Conclusion

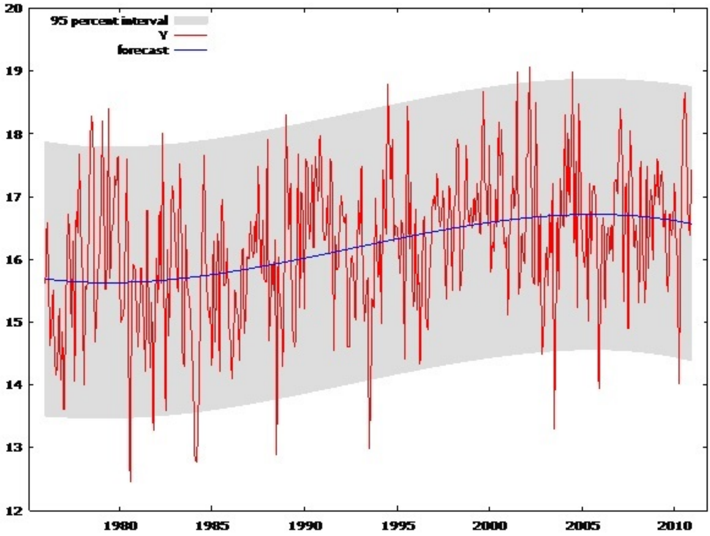
- We have the following model:

$$\underbrace{Y_t}_{\approx X_t - s_t} = \underbrace{15.6925 - 0.00309561 * t + 4.15374 * 10^{-05} * t^2 - 6.95124 * 10^{-08} * t^3}_{\approx m_t} + Z_t.$$

- But from the table we see a_1 is not significant since p-value=$0.4836 > 0.05$. Similarly a_2 and a_3 are insignifficant.

Because of that, this model is preliminary.

The data and the full model



Selection of variables to remove

- We see that we have three insignificant variables;
- We consider a simplification of the model, remembering that the variable with a higher p-value has the priority to remove.
- Using the Wald test we verify the significance of group of variables that are unilaterally insignificant: (VERTE)

Selection of variables to remove (Wald tests)

- **Significance of all variables together:** test $H_0 : a_1 = a_2 = a_3 = 0$ against H_1 : at least one of a_1, a_2 or a_3 is not zero:

  $$F(3, 416) = 20.1917, \quad p - value = 3.10034 * 10^{-12} < 0.05,$$

  we reject the hypothesis a_1, a_2 and a_3 are insignificant together. We cannot remove all variables t, t^2 and t^2 together;
- **Significance of two variables with greatest p-values together:** test $H_0 : a_1 = a_2 = 0$ against H_1 : at least one of a_1 or a_2 is not zero:

  $$F(2, 416) = 8.74472, \quad p - value = 0.000190493 < 0.05,$$

  we reject the hypothesis a_1 and a_2 are insignificant together. We cannot remove both variables t and t^2 together;

As a result, we can reduce the model by removing only one variable, that with the greatest p-value. We remove the variable t.

The reduced model (estimation)

- Because a_1 is not significant, we can consider later the reduced model (without t):

$$Y_t = a_0 + a_1 * t^2 + a_2 * t^3 + Z_t$$

- The results are reported in the next table.

The reduced model (estimation)

	coefficient	std. error=$S_{\hat{a}_i}$	t-ratio= $\frac{\hat{a}_i}{S_{\hat{a}_i}}$	p-value
a_0	15.5617	0.106602	146.0	0.0000
a_1 (by t^2)	$2.49997 * 10^{-5}$	$6.06002 * 10^{-6}$	4.125	$4.47 * 10^{-5}$
a_2 (by t^3)	$-4.50605 * 10^{-08}$	$1.51530 * 10^{-8}$	-2.974	0.0031

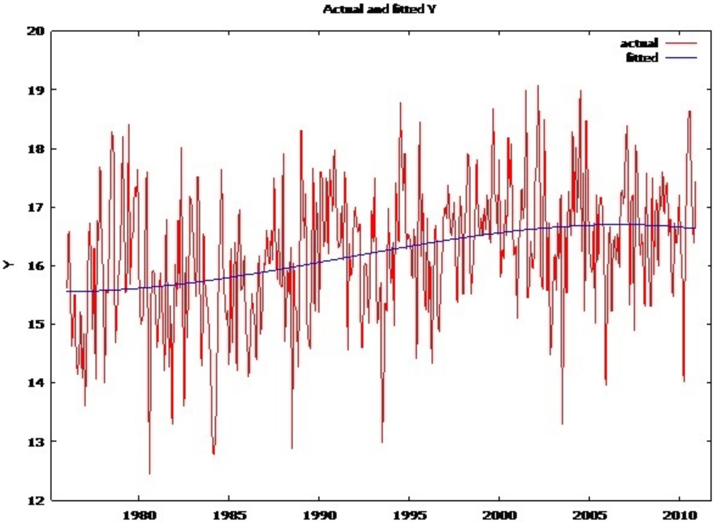
Conclusion

All of the coefficients are significant, hence the following model cannot be reduced anymore:

$$Y_t = 15.5617 + 2.49997 * 10^{-5} t^2 - 4.50605 * 10^{-8} t^3 + Z_t.$$

The reduced model (estimation)

The picture of the data and the regression polynomial is below.



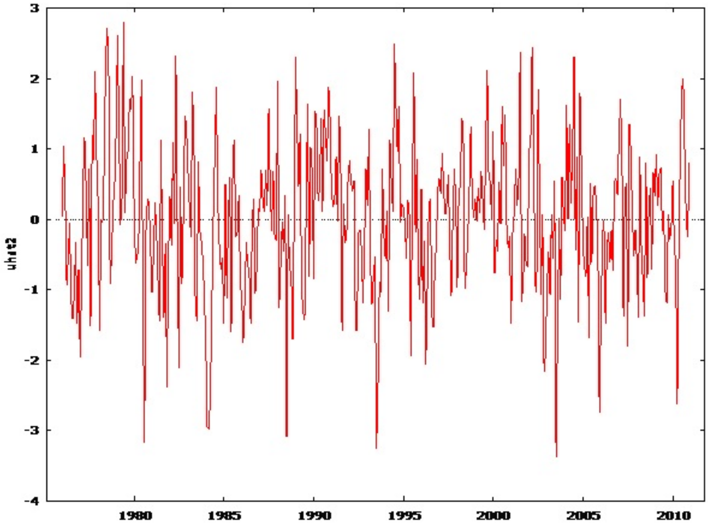
Residual analysis

We analyze the residuals from the reduced model.

$$\hat{Z}_t = Y_t - 15.5617 - 2.49997 * 10^{-5} t^2 + 4.50605 * 10^{-8} t^3$$

The plot of residuals is on the next page.

Residual analysis

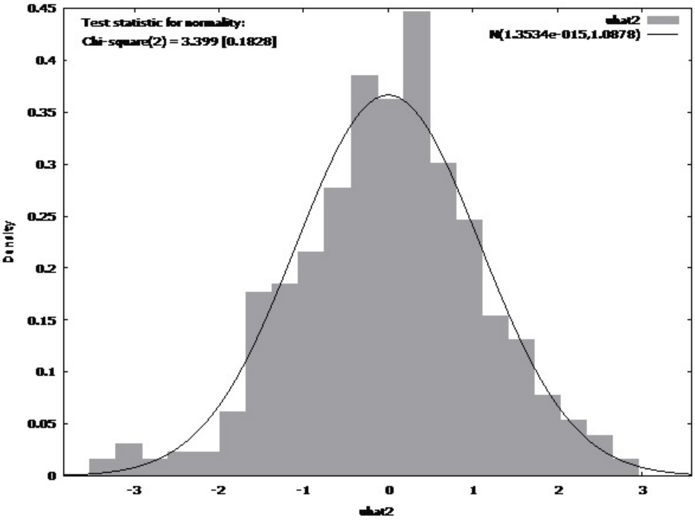


Residual analysis

We verify the normality of residuals.

- Graphically
  - Histogram
  - QQ-plot
- Formally
  - Tests of chi squared, Jarque-Bera, Shapiro-Wilk, Lillefors, Doornik Hansen.

Residual analysis - normality

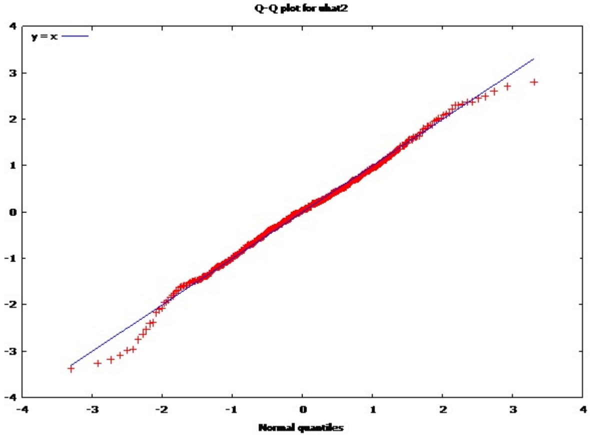


Residual analysis - normality

We interpretate the previous plot

- The test of chi- squared has p-value=$0.1828 > 0.05$ hence this test accepts the hypothesis of normality;
- The histogram looks visually like a normal density.
- The next picture shows the QQ-plot of residuals.

Residual analysis - normality



Residual analysis - normality

We interpretate the QQ plot as follows

- The QQ-plot lies on the diagonal $y = x$;
- It confirms that the datas distribution is normal $\mathcal{N}(0, 1.0878)$ (see the histogram).
- The next picture shows the QQ-plot of residuals.

Conclusion

By the plots of histogram and the so called QQ-plot, we still believe that Z_t has normal distribution. The *chi-square test* confirms the hypothesis.

Residual analysis - tests of normality

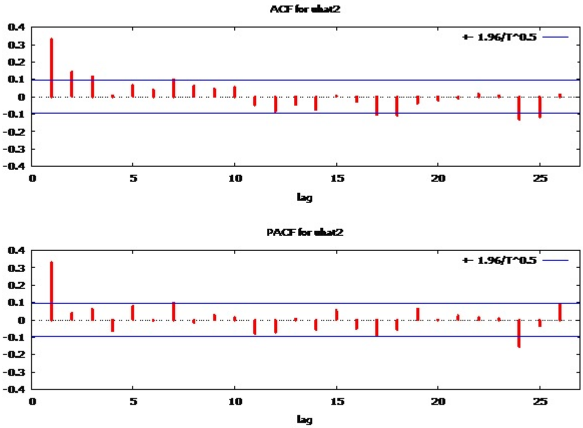
Name of the test	Test statistic	p-value
Doornik-Hansen	$= 3.39879$	$= 0.182794$
Shapiro-Wilk	0.993773	0.0823146
Lilliefors	0.0321819	$= 0.35$
Jarque-Bera	3.53968	0.17036

Conclusion

Since p-value of all tests are greater than 0.05 hence all of the tests above confirm the hypothesis that the residuals have normal distribution. We have eventually verified the hypothesis that Z_t has a normal distribution.

ARMA model of residuals - correlogram

We fit the model by ARMA(p,q) to Z_t. The correlogram is in the picture.



ARMA model of residuals - correlogram

Interpretation:

- The model is ARMA(1,3) or its simplest modifications, ARMA(1,2),ARMA(1,1), MA(3), MA(2), MA(1) or white noise;
- The calculus in the next slide. We start from the most complex model ARMA(1,3) and we simplify the models if some of the coefficients are insignificant.

ARMA model of residuals - estimation of parameters

	coefficient	p-value
ϕ_1	-0.623617	$0.0503 > 0.05$
θ_1	0.950506	$0.0028 < 0.05$
θ_2	0.307579	$0.0057 < 0.05$
θ_3	0.171421	$0.0009 < 0.05$

All but the coefficient multiplied by ϕ_1 are significant. This model for residuals can be reduced to MA(3).

ARMA model of residuals - estimation of parameters

	coefficient	p-value
θ_1	0.332414	$1.25 * 10^{-11} < 0.05$
θ_2	0.115087	$0.0295 < 0.05$
θ_3	0.144343	$0.0038 < 0.05$

Conclusion

All coefficients are significant since p-values are below 0.05. By this table we conclude that the following model MA(3) of residuals is final:

$$Z_t = \epsilon_t - 0.332414\epsilon_{t-1} - 0.115087 * \epsilon_{t-2} - 0.144343 * \epsilon_{t-3}.$$

ARMA model of residuals - estimation of parameters

The model for Z_t is the following:

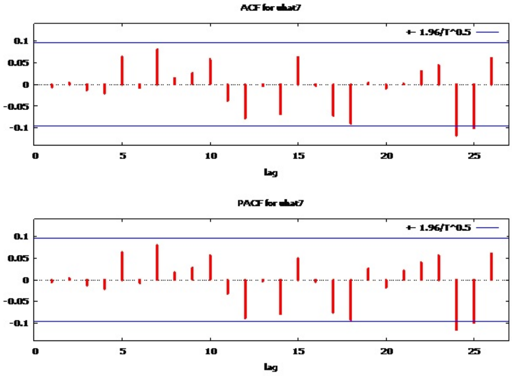
$$Z_t = \epsilon_t - 0.332414\epsilon_{t-1} - 0.115087 * \epsilon_{t-2} - 0.144343 * \epsilon_{t-3}.$$

The final purpose is to verify whether the residuals computed recursively as follows:

$$\epsilon_t = 0.332414\epsilon_{t-1} + 0.115087 * \epsilon_{t-2} + 0.144343 * \epsilon_{t-3} + Z_t$$

form the white noise (with some initial values $\hat{\epsilon}_{-2}, \hat{\epsilon}_{-1}$ and $\hat{\epsilon}_0$). The correlogram for ϵ_t are in the next slide.

ARMA model of residuals ϵ_t



Conclusion

The correlogram confirms that the ϵ_t forms a white noise since ACF and PACF are all not significant.

ARMA model of residuals $\hat{\epsilon}_t$

- To be sure that the model fits good to the data we perform the ***Ljung-Box test*** which tests the joint significance of ACF-s:

$$H_0 : \rho_1 = \rho_2 = \ldots = \rho_h = 0 \quad VS \quad H_1 : \exists_{j=1,2,\ldots,h} \rho_j = 0$$

  where h is not to much not to low. For example, we can take $h \approx 2\sqrt{n} = 2\sqrt{432} \approx 42$.
- The test statistic

$$LB = n(n+2) \sum_{i=1}^{h} \frac{\hat{\rho}_i^2}{n-i} \quad \hat{\rho}_i = ACF_i \text{ (do you remember?).}$$

- Under the hypothesis H_0 the test statistic has the distribution $\chi^2(h)$ (chi squared with $h - p - q$ degrees of freedom, where p and q are parameters of ARMA(p,q)).

ARMA model of residuals $\hat{\epsilon}_t$

The last obstacle for the positive verification of this model is Ljung-Box test.

- We perform this test for $h = 2\sqrt{n} = 2\sqrt{432} \approx 42$.
- We test $H_0 : \rho_1 = \rho_2 = \ldots = \rho_{42} = 0$ against H_1 that the hypothesis H_0 is false.
- The calculations yield: with $h = 42$, $p = 0$ and $q = 3$

$$LB = 43.0975, \quad p{-}value = P(\chi^2_{39} > 43.0975) = 0.3003 > 0.05.$$

Conclusion

The test Ljung-Box accepts the hypothesis H_0, and hence confirms the hypothesis ϵ_t is a white noise. The model is positively verified at all.

Summary

The classical model with

- trend with a 3rd degree polynomial;
- a seasonal component;
- with random element MA(3)

fits to the months temperature in Tokyo during 1976-2010.

Summary

$$X_t = s_t + 15.5617 + 2.49997 * 10^{-5} t^2 - 4.50605 * 10^{-8} t^3 + Z_t$$

where Z_t is ARMA(3,1) process in the form

$$Z_t = \epsilon_t - 0.332414\epsilon_{t-1} - 0.115087 * \epsilon_{t-2} - 0.144343 * \epsilon_{t-3}.$$

and s_t a periodical sequence defined as follows VERTE:

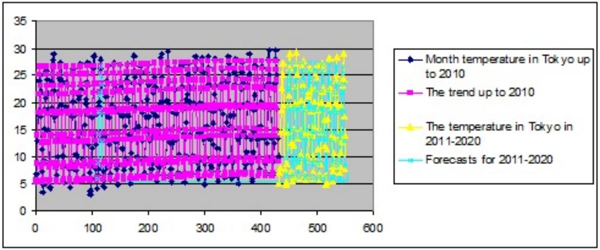
Summary

$$s_t = \begin{cases} -10,2106 & t = 1 \bmod 12 & \text{(for January)} \\ -9,8005 & t = 2 \bmod 12 & \text{(for Febrary)} \\ -6,8775 & t = 3 \bmod 12 & \text{(for March)} \\ -1,6242 & t = 4 \bmod 12 & \text{(for April)} \\ 2,6889 & t = 5 \bmod 12 & \text{(for May)} \\ 5,9917 & t = 6 \bmod 12 & \text{(for June)} \\ 9,5087 & t = 7 \bmod 12 & \text{(for July)} \\ 10,9482 & t = 8 \bmod 12 & \text{(for August)} \\ 7,525 & t = 9 \bmod 12 & \text{(for September)} \\ 2,2756 & t = 10 \bmod 12 & \text{(for October)} \\ -2,8812 & t = 11 \bmod 12 & \text{(for November)} \\ -7,5442 & t = 0 \bmod 12 & \text{(for December)} \end{cases}$$

Exactness of forecasts (Out of competition)

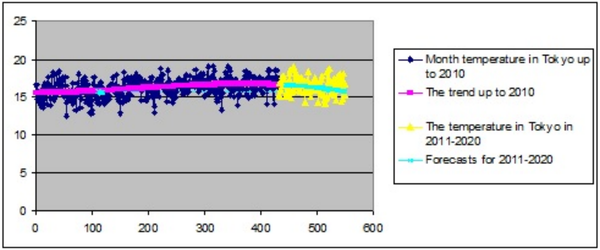
- The procedure is not the subject for your final task.
- By the curiosity, I will present the forecast for 2011-2020 together with the future data in Tokyo.
- The next slide shows the temperature in Tokyo in "testing periods" 1876-2010 and in the "future periods" 2011-2020 with the forecasts.

Exactness of forecasts



The picture above is not illustrative because the seasonality has dominating influence for forming the data. Because of that, I present the data after removing the seasonality.

Exactness of forecasts



At a glance,

- We have expected the cooling trend in 2011-2020;
- But the real trend cools down slower than the predicted.

Exactness of forecasts

- We use later the *ex-post* measures for comparing forecasts from January 2011 to December 2020.
- Let τ be the index of month in this period (i.e. $\tau = 1$ is January 2011, $\tau = 2$ is February 2011 etc.).
- We have $120 = 12 * 10$ months in the period 2011-2020. Let X_τ be the real temperature in the month τ and $\hat{X}_\tau$ its forecast.

We have the following measures of exactness of forecasts

- **Mean Error:**

$$ME = \frac{1}{120}\sum_{\tau=1}^{120}(X_\tau - \hat{X}_\tau);$$

- **Mean Absolute Error:**

$$MAE = \frac{1}{120}\sum_{\tau=1}^{120}|X_\tau - \hat{X}_\tau|;$$

- **Mean Absolute Percentage Error:**

$$MAPE = \frac{1}{120}\sum_{\tau=1}^{120}\frac{|X_\tau - \hat{X}_\tau|}{|X_\tau|} * 100\%;$$

- **Root Mean Squared Error:**

$$RMSE = \sqrt{\frac{1}{120}\sum_{\tau=1}^{120}(X_\tau - \hat{X}_\tau)^2}.$$

Exactness of forecasts

The exactness of the forecasts are reported in the table below:

ME	0.293	The forecast with underflow.
MAE	0.9383	This is average absolute error.
MAPE	5.64%	Mean relative average error is 5.64%.
RMSE	1.142	This is the standard error.

Exactness of forecasts

Very interesting *ex-post* measure is the Teil index:

$$I^2 = \frac{\sum_{\tau=1}^{120} (X_\tau - \hat{X}_\tau)^2}{\sum_{\tau=1}^{120} \hat{X}_\tau^2}$$

and its decomposition on source of errors

$$I^2 = I_1^2 + I_2^2 + I_3^3$$

(VERTE)

where

$$I_1^2 = \frac{\left(\bar{X} - \bar{\hat{X}}\right)^2}{\frac{1}{120}\sum_{\tau=1}^{120} X_\tau^2} \quad I_2^2 = \frac{\left(SE_X - SE_{\bar{\hat{X}}}\right)^2}{\frac{1}{120}\sum_{\tau=1}^{120} X_\tau^2},$$

and

$$I_3^2 = I^2 - I_1^2 - I_2^2.$$

Here

$$\bar{X} = \frac{1}{120}\sum_{\tau=1}^{120} X_\tau, \quad \bar{\hat{X}} = \frac{1}{120}\sum_{\tau=1}^{120} \hat{X}_\tau \quad (\text{note } \hat{X} = \hat{\hat{X}})$$

are mean averages of future values and their expectations respectively and

$$SE_X = \frac{1}{120}\sum_{\tau=1}^{120}(X_\tau - \hat{X})^2, \quad SE_{\bar{\hat{X}}} = \frac{1}{120}\sum_{\tau=1}^{120}(\hat{X}_\tau - \bar{\hat{X}})^2.$$

Exactness of forecasts

The following values indicate shares of corresponding sources of errors in explaining the prediction error:

- The following value indicates the share of the difference between the mean averages of forecasts and exact values:

$$\tilde{I}_1^2 = \frac{I_1^2}{I^2} 100\%;$$

- The following value indicates the share of the difference between the standard errors (varieties) of forecasts and exact values:

$$\tilde{I}_2^2 = \frac{I_2^2}{I^2} 100\%;$$

- The following value indicates the share of other source of prediction error (like incomplete correlation between forecasts and exact values):

$$\tilde{I}_3^2 = \frac{I_3^2}{I^2} 100\%;$$

Exactness of forecasts

Remark

Since the source of error prediction hidden in $\tilde{I}_3^2$ is out of control of the statistician, the ideal model has the following distribution:

$$\tilde{I}_1^2 = 0 \quad \tilde{I}_2^2 = 0 \quad \text{and} \quad \tilde{I}_3^2 = 100\%.$$

In other words, the good statistical model should have under control the difference between mean averages and the difference between standard errors (or varieties) of both forecasts and the exact values.

Exactness of forecasts

In case of the month temperature in Tokyo, we report the shares (up to rounding to 2 space after comma) of the sources of errors in the table below:

Source of error	Value	Share
Different means	$I_1^2 = 0.26 * 10^{-3}$	$\tilde{I}_1^2 = 6.59\%$
Different varieties	$I_2^2 = 0.39 * 10^{-3}$	$\tilde{I}_2^2 = 9.87\%$
Other sources	$I_3^2 = 3.31 * 10^{-3}$	$\tilde{I}_3^2 = 83.54\%$
Total	$I^2 = 3.96 * 10^{-3}$	100%

Remark

The distribution of the Theil index shows that the model controls the difference between mean averages and varieties of forecasts and exact future values in 83.54%. The distinct averages is a 6.59% source of error prediction, and distinct standard errors is a 6.59% source of error prediction.